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The Anchoring Effect, Algorithmic Fairness, and the Limits of Information Transparency for Emotion Artificial Intelligence

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Abstract. Emotion artificial intelligence (AI) or emotion recognition AI may systematically vary in its recognition of facial expressions and emotions across demographic groups, creating inconsistencies and disparities in its scoring. This paper explores the extent to which individuals can compensate for these disparities and inconsistencies in emotion AI considering two opposing factors; although humans evolved to recognize emotions, particularly happiness, they are also subject to cognitive biases, such as the anchoring effect. To help understand these dynamics, this study tasks three commercially available emotion AIs and a group of human labelers to identify emotions from faces in two image data sets. The scores generated by emotion AI and human labelers are examined for inference inconsistencies (i.e., misalignment between facial expression and emotion label). The human labelers are also provided with the emotion AI's scores and with measures of its scoring fairness (or lack thereof). We observe that even when human labelers are operating in this context of information transparency, they may still rely on the emotion AI's scores, perpetuating its inconsistencies. Several findings emerge from this study. First, the anchoring effect appears to be moderated by the type of inference inconsistency and is weaker for easier emotion recognition tasks. Second, when human labelers are provided with information transparency regarding the emotion AI's fairness, the effect is not uniform across emotions. Also, there is no evidence that information transparency leads to the selective anchoring necessary to offset emotion AI disparities; in fact, some evidence suggests that information transparency increases human inference inconsistencies. Lastly, the different models of emotion AI are highly inconsistent in their scores, raising doubts about emotion AI more generally. Collectively, these findings provide evidence of the potential limitations of addressing algorithmic bias through individual decisions, even when those individuals are supported with information transparency.

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1. Introduction

Emotion recognition artificial intelligence (AI; emotion AI) is a growing subset of facial analysis used to automatically recognize emotions. It is already used to screen job applicants and determine threats (Thomas 2018, Harwell 2019, Telford 2019); is expanding into mental health apps, call centers, and advertising feedback (Somers 2019); and is being incorporated into virtual meeting software to measure participants' engagement (O'Leary 2022).

Despite its benefits, emotion AI may exhibit disparities in the quality of its recognition of facial expressions and emotions based on the subject's demographic characteristics (Xu et al. 2020), suggesting algorithmic bias. Organizations may attempt to counteract such algorithmic bias in multiple, complementary ways; they may arm their end users to counter bias through individual choices, adopt audited or fairness-aware algorithms, and/or exert public pressure on algorithm developers to debias their algorithms (Raji and Buolamwini 2019). The first of these tools, end user oversight, is a valuable aid for emotion AI because humans have evolved to recognize emotions. However, researchers in behavioral economics and applied psychology note that because of the anchoring effect, any numeric input that a human evaluator sees may influence their judgment (Tversky and Kahneman 1974, Adomavicius et al. 2013). Organizations that rely on end users to counteract algorithmic bias may find that these end users have their own cognitive biases and are, therefore, influenced by emotion AI outputs, so disclosing information about AI fairness measures may not yield higher-

quality individual choices. Thus, emotion AI is a good context for studying the interplay of algorithmic fairness, information transparency, and the anchoring effect.

This research examines whether individuals can counteract algorithmic inconsistencies and disparities in emotion AI and yield consistent scores. To this end, we conduct a study on two facial image data sets: one data set of National Basketball Association (NBA) players and another from the Balanced Faces in the Wild (BFW) data set, which was designed for image recognition (Robinson et al. 2020). The emotions in those faces are scored using three commercially available emotion AIs that may exhibit racial disparities and inference inconsistencies (i.e., misalignment of the facial expression and the emotion score). Then, we conduct an experiment in which human labelers provided with varying information about the emotion AI, including the scores it generated and its known level of fairness or bias, score the emotions in those same faces.

This study yields several interesting findings. As expected, there is consistent evidence of the anchoring effect and some evidence of a selective anchoring effect. First, the anchoring effect appears to be moderated by the type of inference inconsistencies in the emotion AI scores such that the effect appears weaker for the easier task of happiness recognition. Second, the effect of information transparency about algorithmic fairness is not uniform across emotions. This result may occur because humans are better at recognizing happiness and rely less on the emotion AI as an anchor. Lastly, there is no evidence that information transparency leads to the selective anchoring needed to offset the emotion AI disparities, and there is limited evidence that information transparency can increase human inference inconsistencies. Collectively, these findings demonstrate that even under conditions of information transparency with regard to algorithmic fairness, individual users may have limited ability to address algorithmic bias through their decisions.

This study makes multiple contributions to the information systems literature. It finds evidence that the anchoring effect is stronger in response to harder tasks (which is also when human oversight is particularly important to remedy AI-based inference inconsistencies), suggesting that cognitive biases could limit the effectiveness of this type of human oversight. It also finds that information transparency is more influential in harder tasks, but it affects the labelers' scores across all faces. Overall, these findings indicate that information transparency regarding algorithmic fairness is not enough to cause individuals to adjust their behavior, and separately, it may also yield lower-quality decisions. Because of the extreme variation in scores, this study raises doubts about the real-world performance and implementation of emotion AI.

2. Theoretical Background

This research is related to four research streams: (1) emotion AI, (2) algorithmic bias, (3) information transparency, and (4) the anchoring effect. We briefly summarize those streams.

2.1. Emotion AI

Emotion AI, a subset of facial recognition and analysis technology, was developed to automatically identify emotions from facial expressions (Benitez-Quiroz et al. 2016, Dehghan et al. 2017). Emotion AI models work by identifying patterns of facial expressions based on human-labeled training data, often using probabilistic predictive models such as neural networks (Cohen et al. 2000). Facial analysis models employ landmark extraction and identify facial attributes, such as the chin, eyes, nose, and mouth, from photos of individuals (Xu et al. 2016). In emotion AI tasks, sets of facial images are labeled with the associated emotions by human labelers. Given enough examples of a particular facial expression with its associated emotion label, an emotion AI model will learn the factors in the facial expression that predict a specific emotion (Cohen et al. 2000). This paper builds on the emotion AI literature by investigating potential disparities in emotion AI based on demographic characteristics, exploring questions of fairness in emotion AI and the potential use of information transparency as a means of mitigating fairness concerns.

2.2. Algorithmic Bias and Fairness

Algorithmic bias occurs when algorithmic failures or inconsistencies are correlated with demographic characteristics. Facial recognition and analysis algorithms, including emotion AI, have shown evidence of algorithmic bias (Buolamwini and Gebru 2018, Xu et al. 2020), such as providing less accurate predictions for darker-skinned faces. Algorithmic bias can have many causes (e.g., human bias in creating the data labels or overrepresentation of one population in the training set) (Torralba and Efros 2011). A multidisciplinary approach has emerged that combines statistical techniques and organizational frameworks to combat algorithmic bias and enhance nondiscriminatory treatment (Hardt et al. 2016, Corbett-Davies et al. 2017, Kleinberg 2018, Kleinberg et al. 2018, Raji and Buolamwini 2019, Turner Lee et al. 2019).

Broadly, nondiscriminatory treatment is understood as fairness. Measures of algorithmic fairness quantify variations in how algorithms handle different groups and/or individuals in different groups; they can thus serve as mechanisms for diagnosing the presence of algorithmic bias. One common measure of algorithmic fairness is demographic parity (DP) (i.e., the difference in the expected value of an outcome variable conditional on demographic characteristics alone).[1] DP is a

commonly used fairness measure because it is easy to calculate and straightforward to explain. However, it is not always appropriate (Hardt et al. 2016, Awwad et al. 2020, Mehrabi et al. 2022, Teodorescu et al. 2022). This can be the case when demographic groups have meaningful underlying variations, so the outcome variable may display significant group-level differences and still adhere to definitions of fairness (Kleinberg 2018). Plus, DP may fail to appropriately consider individuals on the decision boundary (Awwad et al. 2020), or it may harm the protected group over a longer time horizon (Fu et al. 2022). Despite these concerns, DP is often used because it aligns with human perceptions of fairness (Srivastava et al. 2019).

As organizations continue to adopt artificial intelligence into their processes and products, procedures to enhance algorithmic fairness are an important consideration in information systems development (Teodorescu et al. 2021). Most statistical techniques to increase algorithmic fairness rely on access to the underlying models; however, commercially available facial recognition systems (such as IBM Watson, Microsoft Cognitive Services, Face++, and Amazon Rekognition) provide client access to pretrained probabilistic models without providing access to the underlying models. Without this access, client organizations cannot see these models' methods (chapter 1 in O'Neil 2016), precluding an analysis of procedural fairness (Morse et al. 2022). Given this limitation and the growing concern about unfair algorithmic behavior (Barocas and Selbst 2016, Buolamwini and Gebru 2018), clients may want to audit these proprietary models for fairness in the outcomes instead of fairness in the procedures.

2.3. Information Transparency Regarding Algorithmic Fairness

Understanding fairness in the outcomes for any specific context requires information transparency regarding the underlying algorithm. Such information transparency involves "the presence of information on an algorithm's structure, procedure, decision rationale, or performance" (Lehmann et al. 2022, p. 3421). Yet, this information transparency, such as details about the training data and classifiers, is often unavailable for commercially available pretrained models (Barocas and Selbst 2016). Instead, such models often undergo outcome-based audits, such as the National Institute of Standards and Technology Facial Recognition Vendor Test, which reports a model's false-positive rate, false-negative rate, and any disparities along demographic characteristics (Grother et al. 2019).

Information transparency around racial disparities in AI outputs can spur designers to change their training data and/or models to encourage algorithmic fairness (Buolamwini and Gebru 2018, Raji and Buolamwini 2019) and galvanize organizations to regularly audit their algorithmic tools for bias (Turner Lee et al. 2019). In general, information transparency may reduce the negative externalities from artificial intelligence (Luca et al. 2016, Rhue and Washington 2020); however, it can also exacerbate disparities by delaying bias to a later point in the process (Mejia and Parker 2021). This paper contributes to the algorithmic fairness and information transparency literatures by studying how information transparency regarding algorithmic fairness affects individual choices.

2.4. Cognitive Bias and the Anchoring Effect

The effectiveness of information transparency on algorithmic fairness depends on the degree to which humans are susceptible to cognitive biases, such as the anchoring effect (Tversky and Kahneman 1974). Emotion AI scores are expected to influence human labelers because any information presented before a judgment is likely to act as an anchor for that judgment (Tversky and Kahneman 1974), even if these anchors are random and uninformative (Tversky and Kahneman 1974, Ariely et al. 2003). In the seminal study of the anchoring effect, subjects were asked to spin a wheel to obtain a number; then, they were asked to estimate the number of countries in the United Nations (Tversky and Kahneman 1974). The researchers found that even though subjects know that the number is randomly generated, they still insufficiently adjust from that anchor point (Tversky and Kahneman 1974), and this basic anchoring effect has been replicated (Brewer and Chapman 2002).

Research on the anchoring effect lies within multiple streams (Adomavicius et al. 2013). One stream discusses the underlying psychological mechanisms (Adomavicius et al. 2013), with the original explanation being insufficient adjustment from the anchor (Tversky and Kahneman 1974). Other theories include numeric priming, where the presence of a number can "prime" the respondent, or scale distortion, which is when people use the anchor to inform the scale for their judgments (Brewer and Chapman 2002, Mochon and Frederick 2013). A related stream of literature examines factors that moderate the anchoring effect, such as familiarity with the decision and trust in the anchor's source (van Exel et al. 2006). This paper expands on the anchoring effect literature by examining how the anchoring effect interacts with the algorithmic fairness literature.

3. Hypothesis Development

3.1. The Anchoring Effect and Inference Inconsistencies

One lens through which to analyze the quality of an AI's emotion scoring is inference inconsistency. In this context, inference inconsistency describes the misalignment between a facial expression and an emotion. For instance, a smile is a nearly universal indicator of

happiness in pictures (Kotchemidova 2005), but a human or an emotion AI may mislabel a smiling face as angry. Inference inconsistencies can occur in two directions: a false-negative inference inconsistency (FNII) or a false-positive inference inconsistency (FPII). FNII denotes a situation in which the model or labeler mislabels a smiling face as nonhappy (i.e., a smile is present, but the emotion is not happy). Conversely, FPII denotes the situation in which a nonsmiling face is mislabeled as happy (e.g., no smile is present, but the emotion is happy). We consider these inference inconsistencies as examples of algorithmic failures on the part of an emotion AI and examples of poor quality on the part of a human labeler.

We would expect humans, when presented with inference inconsistencies generated by AI, to provide higher scores in the appropriate emotion because humans generally excel at emotion recognition. Although humans should produce scores that offset inference inconsistencies in both positive and negative directions, their emotion recognition ability is not uniform for all emotions. Specifically, humans more easily recognize happy emotions in facial expressions (Johnston et al. 2001, Leppänen and Hietanen 2004, Bänziger et al. 2009, Vaish et al. 2009) and are better at recognizing happy faces (Kirita and Endo 1995). If humans' emotion recognition abilities were truly consistent, we would expect them to label faces with FNII as happier and faces with FPII as less happy. However, given that people are better at recognizing happiness, human labelers may be more successful at compensating for FNII than FPII; therefore, we expect to see a weaker anchoring effect on happy emotions and a stronger anchoring effect other emotions.

Hypothesis 1a. *The anchoring effect is weaker in the case of inference inconsistencies for easier tasks, such as mislabeled smiling faces (FNII).*

Hypothesis 1b. *The anchoring effect is stronger in the case of inference inconsistencies for harder tasks, such as mislabeled nonsmiling faces (FPII).*

3.2. Information Transparency About Algorithmic Fairness

The anchor effect may be moderated by the anchor's source, and the framing of information can influence how consumers incorporate that information into their judgments (Adjerid et al. 2018, Bassellier and Ramaprasad 2023). This means that higher trust in an anchor's source is more likely to yield judgments consistent with the anchoring effect (van Exel et al. 2006, Furnham and Boo 2011). Humans tend to trust technology and artificial intelligence (Eslami et al. 2018) in a phenomenon known as *algorithmic appreciation* (Logg et al. 2019); specifically, studies have found that people trust algorithmic advice at least as much as human advice (Gunaratne et al. 2018, Bogert et al. 2021). Thus, in the context of emotion AI and from the perspective of a human observer, information transparency about fairness measures could highlight the fact that the anchors (emotion AI scores) may be inaccurate, undermine confidence in the anchors, and weaken the anchoring effect (van Exel et al. 2006, Furnham and Boo 2011). However, in uncertain circumstances, human labelers' cognitive biases have a stronger effect, so they are more likely to rely on anchors (van Exel et al. 2006, Furnham and Boo 2011). Labelers are particularly likely to rely on anchors for more difficult tasks, for unfamiliar situations, and/or when the labeling task involves marginalized people (Nowicki et al. 1998, van Exel et al. 2006). Therefore, we expect that information transparency will lower the anchor effect for the easier task (scoring happiness) and strengthen it for the harder task (scoring anger). We hypothesize the following.

Hypothesis 2a. *Information transparency about AI fairness will shape labelers' scores less in the easier task.*

Hypothesis 2b. *Information transparency about AI fairness will shape labelers' scores more in the harder task.*

3.3. Information Transparency and the Anchoring Effect

Information transparency regarding AI fairness could encourage labelers to reflect on algorithmic fairness, reconsider their proposed scores, and make choices intended to offset any disparities. Human emotion recognition varies by racial groups (Kilbride and Yarczower 1983); people are less accurate at recognizing emotions from marginalized groups, such as African Americans (Nowicki et al. 1998), and may be more likely to assign anger to their facial expressions (Halberstadt et al. 2022). Given the cultural association of anger with marginalized groups, such as African Americans, this study focuses on happiness, the most easily recognized emotion, and anger, the emotion most likely to interact with racial characteristics. We expect that information transparency would cause the anchoring effect to be moderated by the face's demographic characteristics, creating a selective anchoring effect.

Ultimately, information transparency could also increase human inference inconsistencies. This is because emotions, demographic characteristics of the face, and information transparency may all shape the labeler's confidence in their task. People tasked with emotion recognition for unfamiliar or traditionally marginalized groups may feel more uncertainty (Kilbride and Yarczower 1983), which would strengthen the anchoring effect (van Exel et al. 2006, Furnham and Boo 2011); however, information transparency regarding algorithmic fairness may undermine trust and weaken the anchoring effect. Thus, information transparency may complicate the labeler's assessment of the emotion AI's trustworthiness, yielding

a higher chance of human inference inconsistencies. We expect the following.

Hypothesis 3a. *The effect of information transparency about AI fairness measures on labelers' scores is moderated by demographic characteristics.*

Hypothesis 3b. *Information transparency about AI fairness measures leads to inconsistency in labelers' scores.*

4. Data

This section discusses this study's tools, variable construction, and data collection procedures.

4.1. Tools and Variable Definitions

This analysis used three different and common emotion AI tools: Face++,[2] Microsoft Azure Face Application Programming Interface (API),[3] and Amazon Rekognition.[4]

4.1.1. Face++. Face++, a Megvii product, was launched in Beijing in 2012 as one of the world's largest facial recognition tools.[5] Face++ analyzes faces for emotion, smile, gender, image quality, and race (before October 2019). For emotion, each face is assigned a value for each of seven emotions: anger, disgust, fear, happiness, neutral, sadness, and surprise. Each face is then labeled based on the emotion with the highest numeric amount for that face. The sum of the seven emotional scores equals 100, which in our study, was rescaled to 10. Face++ identifies the degree to which a face is smiling, and this attribute is distinct from the assessment of happiness.

4.1.2. Microsoft Azure Cognitive Services. Microsoft's Azure Face API provides facial recognition services to numerous large companies (e.g., Uber) and to researchers (e.g., Choudhury et al. 2019). It analyzes faces based on emotion, smiling, and noise level. Each face is scored on eight emotional dimensions: anger, contempt, disgust, fear, happiness, neutral, sadness, and surprise. For each face, the scores for all the emotions sum to one. The emotion with the highest numeric amount becomes the emotion label for that face. In addition to rating emotions, Microsoft rates each face's smile between zero and one, and the smile score is the same as that face's happiness score. In order to keep the scores on the same scale, we rescaled the Microsoft scores so that the highest possible value for smile and happiness is 10.

4.1.3. Amazon Rekognition. Rekognition, Amazon's image recognition service, is one of the most widely used facial recognition services. Amazon Rekognition's face detection and analysis API scored faces on several emotions: calm, disgusted, angry, sad, happy, surprised, and confused. The emotion with the highest numeric amount is the emotion label for that face. These scores also sum to 100 for each face and were rescaled so that the highest possible value for happiness is 10. Amazon provided a binary indicator for whether the facial expression contains a smile and the confidence interval for that smile prediction.

4.2. Image Data Sets

To address concerns about data set composition, this paper uses two image data sets.

4.2.1. NBA Data Set. The first data set is a collection of professional images of 471 NBA players from the 2016–2017 season that were scraped from Basketball Reference.[6] These pictures were taken in a professional context with relatively standard poses. All players faced the camera head-on, a preferable position for accurate facial analysis. These players were relatively homogenous in their age, gender, and physicality but fell mainly into two racial categories: African American (75%) or White (25%). These pictures were scored in 2017 using all three emotion AI tools (Face++, Microsoft, and Amazon).

4.2.2. Balanced Faces in the Wild Data Set. The second data set is the BFW data set (Robinson et al. 2020). This data set was designed to create a gender and racially balanced data set for facial recognition and was based on the VGGFace2 data set (Cao et al. 2019). BFW provided 100 faces, with four races (East Asian, African American, South Asian, and White) and two genders (male and female) represented. For each face, they provided multiple images. To narrow down the data set, we selected the largest image (by size) for each face. This resulted in 100 pictures per category and a total of 800 pictures. There were abnormalities in 14 in the pictures, leaving a total of 786 pictures in the final data set. The pictures were scored in 2020 using all three emotion AI tools.

4.3. Emotion AI Model Scores: Disparities

The means and standard errors of the emotion AI scores for each data set are displayed in Table 1. There was slightly more variation in average emotion AI scores within the BFW data set, which is a compilation of pictures in different poses and facial expressions, as compared with the NBA data set, which is a set of professional headshots. However, for both the NBA and BFW data sets, the scores varied dramatically between emotion AI models. Emotion AI models also differed in their propensity for racial disparities. Microsoft exhibited low racial disparities in emotion scores. Face++ exhibited the widest racial disparities in emotion scores in the NBA data set, consistent with prior research in which Face++ had one of the highest disparities in accuracy rates for gender identification of people with darker skin (Buolamwini and Gebru 2018). Despite low

Table 1. Emotion Recognition Models' Scaled Emotion Scores for Happiness and Anger

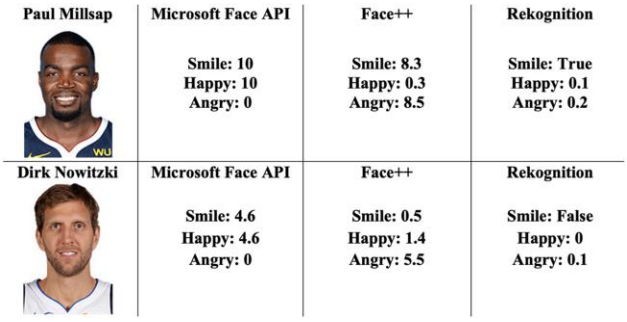
		Microsoft		Amazon		Face++	
Group	*N*	*Happy* Mean (SE)	*Angry* Mean (SE)	*Happy* Mean (SE)	*Angry* Mean (SE)	*Happy* Mean (SE)	*Angry* Mean (SE)
NBA data set							
Overall	471	6.26 (0.21)	0.01 (0.01)	1.65 (0.13)	1.52 (0.12)	5.46 (0.18)	0.42 (0.04)
White	117	6.73 (0.41)	0.02 (0.02)	1.89 (0.28)	1.53 (0.28)	6.17 (0.37)	0.21 (0.06)
AA	354	6.11 (0.24)	0.01 (0.00)	1.57 (0.15)	1.52 (0.14)	5.23 (0.21)	0.49 (0.05)
BFW data set							
Overall	786	4.69 (0.16)	0.13 (0.03)	0.14 (0.01)	0.435 (0.02)	4.37 (0.16)	0.39 (0.05)
White	198	4.98 (0.33)	0.20 (0.07)	0.12 (0.01)	0.43 (0.04)	4.69 (0.33)	0.38 (0.11)
E. Asian	196	3.94 (0.32)	0.03 (0.02)	0.11 (0.01)	0.36 (0.03)	3.60 (0.31)	0.06 (0.01)
AA	195	5.32 (0.34)	0.09 (0.03)	0.14 (0.01)	0.44 (0.04)	5.14 (0.33)	0.39 (0.09)
S. Asian	197	4.55 (0.32)	0.21 (0.08)	0.17 (0.01)	0.50 (0.04)	4.07 (0.31)	0.71 (0.16)

Note. AA, African American; E. Asian, East Asian; S. Asian, South Asian; SE, standard error.

racial disparities, Amazon's scores for happiness did not align with those of the other emotion AI models.

The three emotion AI models produced notably different scores for the same faces. Figure 1 illustrates an example of this score divergence. For the picture in the upper row, Paul Millsap, Microsoft produced a 10 of 10 for happiness and 0 for anger. Face++ identified Millsap as both smiling (8.3 of 10) and angry (8.5 of 10). Amazon labeled Paul Millsap's emotions as low on both happiness and anger (<1 of 10). The second picture, Dirk Nowitzki, reveals a similar story. Microsoft labeled his emotions as 4.6 for happiness, but Face++ labeled his facial expression as 5.5 for anger, and Amazon labeled his expression as low on both happiness and anger. A more detailed examination of the disparities between the emotion AI models is provided in the online appendix.

Figure 1. (Color online) Example Pictures



4.4. Inference Inconsistencies

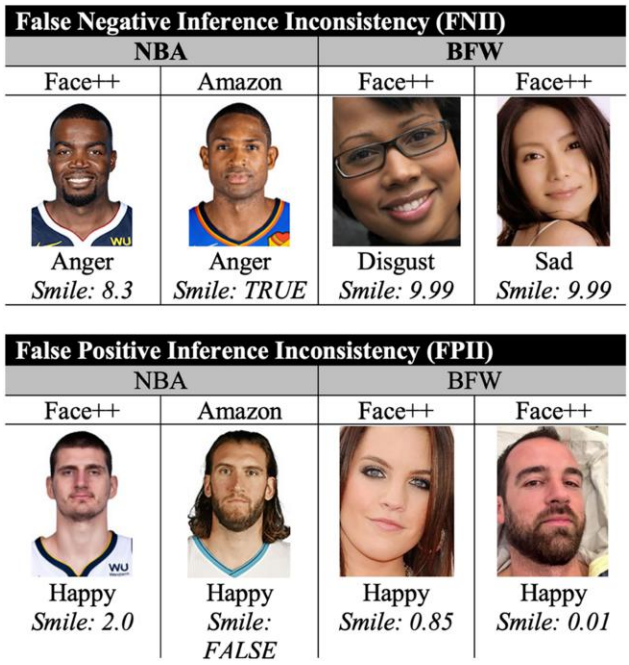
The emotion AI models demonstrated inference inconsistency in two directions: a false negative for happiness (i.e., a smiling face and a nonhappy emotion) or a false positive for happiness (i.e., a nonsmiling face and a happy emotion). Because smiling and nonsmiling faces can both be deemed neutral, neutral emotion was never inconsistent. Figure 2 shows examples of internal inconsistencies from both the NBA and BFW data sets and for Face++ and Amazon emotion AI models. Microsoft did not have any inference inconsistencies because it reported identical scores for smiling and happy. Amazon identified all faces in the BFW data set as calm or neutral, so Amazon had no inference inconsistencies in the BFW data set.

5. Empirical Analysis

5.1. Experimental Design and Summary Statistics

In our nested experimental design, we asked human labelers to identity emotions from the facial expressions in the two data sets. There are two experimental conditions. (1) In the anchor condition, the labeler was exposed to the scores generated by one of the emotion Ais, and (2) in the transparency condition, the labeler was exposed to the scores generated by one of the emotion AIs and also received information about the demographic parity of that emotion AI. Labelers in the control group were asked to rate a face without any information about emotion AI scores. In the anchor

Figure 2. (Color online) Examples of FNII and FPII



condition, labelers received an emotion AI score for a particular face from one of the three models before scoring the facial expression. The face was labeled with the statement, "An artificial intelligence (AI) scored this face as <score> out of 10," where <score> is the emotion AI score for that face for a given model. In the screen shown to the labelers, the emotion AI models were anonymized to remove any brand influence. For the anchor plus information transparency dimension, labelers received the emotion AI score along with the average racial disparity for each emotion. The experiment differed slightly for the NBA and BFW data sets.

5.1.1. NBA Data Set. Each picture in the NBA data set received a score from three unique labelers. For the Anchor conditions, labelers could be exposed to one of the three platforms (Microsoft, Amazon, or Face++) or none (control). Labelers scored the facial expressions on a 0–10 scale for happiness, anger, and sadness. Figure 3 shows a sample of the control, anchor, and transparency conditions.

Table 2 shows the mean scores and the standard errors across the experimental conditions. The anchors do appear to influence the labelers' mean scores. For instance, labelers exposed to the Microsoft anchors rated the players' happiness higher on average than the labelers with the Face++ and Amazon anchors. This is consistent with the anchoring effect.

Across all conditions, exposure to emotion AI scores reduced the anger scores. On average, the control group labeled the facial expressions as 1.25 angry, whereas the anchor groups all labeled the facial expressions lower. Emotion AI appears to engender different scores for the same set of faces, consistent with the anchoring effect.

5.1.2. BFW Data Set. For the BFW data set, the anchor dimension included exposure to either Microsoft's or Amazon's emotion AI scores or to none (control). Labelers then judged the facial expression on a 0–10 scale for happiness, anger, and neutrality. Each picture received a score from two unique labelers, and we report the mean score across the labelers. We chose to focus on Microsoft and Amazon because these two companies had the lowest disparities and were the most likely to be used by companies in 2020. Table 3 shows the summary statistics for the labelers' scores.

5.2. The Anchoring Effect

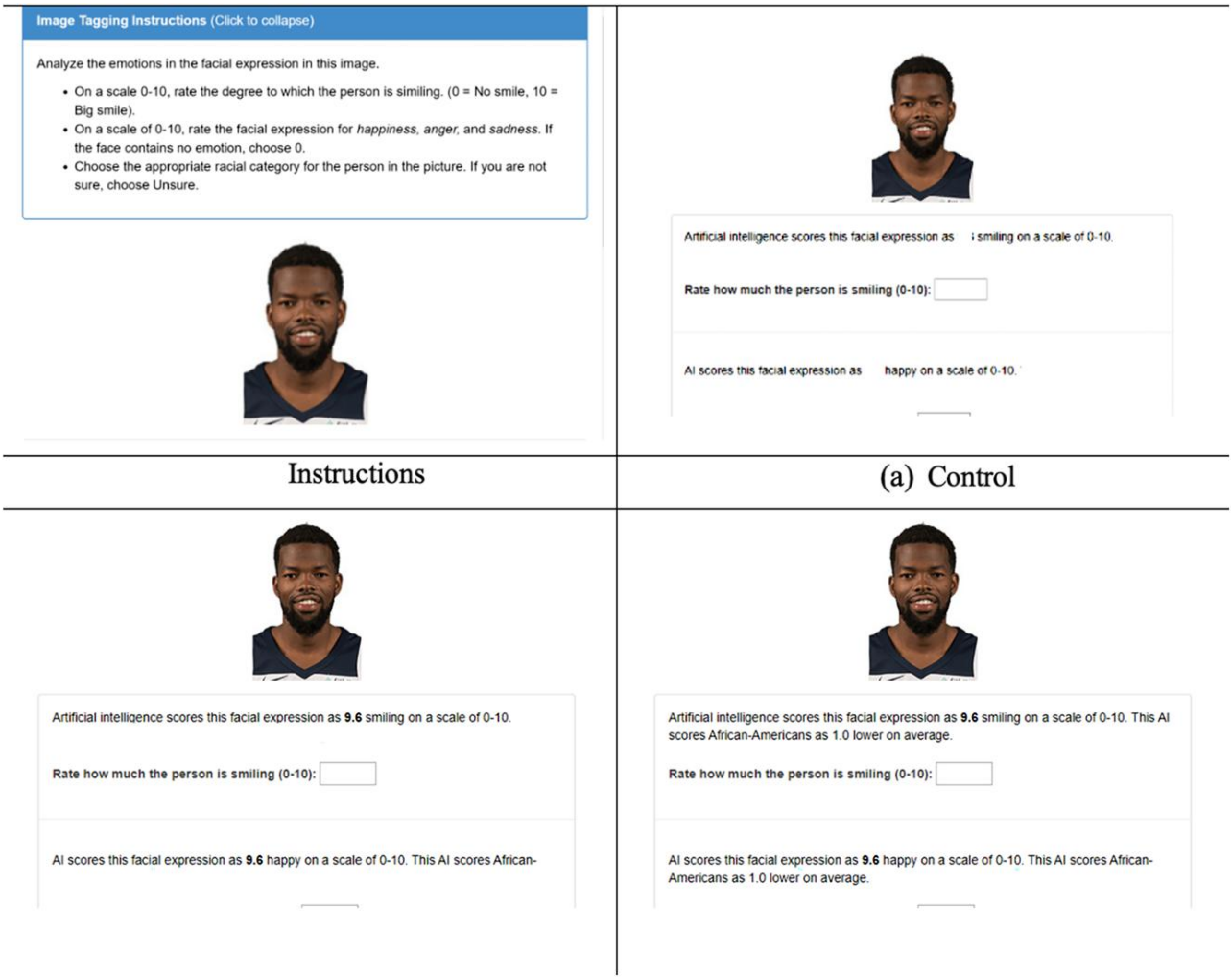
To learn how labelers' scores are shaped by emotion AI scores, we compared the scores for the same subject across different anchors. We used a mixed-level model with random effects, from the lme4 R package (Bates et al. 2015), to control for differences in each subject's facial expression. The following model specifications were used for the NBA and BFW data sets:

$$LabelerScores_{ij} = \alpha + \beta_1 AIScore_{ij} + \beta_2 Race_i + \beta_3 AIScore_j \times Race_i + \epsilon_i + \epsilon_j,$$

where *LabelerScores* are the scores that labelers assign to the facial expression for subject i in condition j, *AIScore* is the emotion AI score for subject i by emotion AI model j, and *Race* is the racial category of subject i. For the NBA data set, *Race* is a binary variable denoting whether subject i is African American. For the BFW data set, *Race* is a set of binary variables denoting whether subject i belongs to one of the four racial groups (White, East Asian, African-American, and South Asian). *Emotion AI Score* × *Race* is the interaction between the two variables. There is a subject-level random effect ϵ_i to control for differences between subjects and an emotion AI random effect ϵ_j to control for systematic differences among the emotion AIs. For this analysis, we exclude the transparency conditions because information about AI fairness would create noise in the analysis.

5.2.1. NBA Data Set. Table 4 provides a baseline for the anchoring effect. As expected, these findings suggest that the labelers' scores are a function of the emotion AI score for both happiness and anger ($\beta = 0.07$ and $\beta = 0.11$, respectively). The results also suggest some selective anchoring, as the labelers are more likely to incorporate the emotion AI scores for anger for African American subjects ($\beta = 0.07$).

Figure 3. (Color online) Sample Conditions for the NBA Data Set



5.2.2. BFW Data Set. Table 5 shows the results. Human labelers were similarly influenced by the emotion AI scores; the labelers' scores for happiness and anger were positively associated with the emotion AI scores for those emotions ($\beta = 0.16$ and $\beta = 0.52$, respectively). There is evidence of selective anchoring; the emotion AI scores were less influential for Asian subjects ($\beta = -0.05$ and $\beta = -0.28$, respectively).

5.2.3. Emotion AI Inference Inconsistencies. The findings provide a baseline to support the expectation of the anchoring effect and the moderating influence of

Table 2. Mean Labelers' Emotion Scores by Condition for the NBA Data Set

Anchor	Bias mitigation	*N*	Happy	Angry	Sad	Smile
Amazon	None	1,413	4.91 (0.15)	0.87 (0.07)	0.74 (0.06)	4.96 (0.16)
	Transparency	1,413	4.98 (0.15)	1.08 (0.06)	1.33 (0.07)	5.05 (0.15)
Face++	None	1,413	4.63 (0.13)	0.87 (0.06)	0.72 (0.05)	4.55 (0.13)
	Transparency	1,413	4.86 (0.14)	0.77 (0.06)	0.89 (0.06)	4.82 (0.15)
Microsoft	None	1,413	5.25 (0.16)	0.63 (0.05)	0.59 (0.05)	5.23 (0.16)
	Transparency	1,413	5.12 (0.16)	0.48 (0.04)	0.62 (0.05)	5.26 (0.16)
Control	None	1,413	4.93 (0.13)	1.25 (0.08)	1.3 (0.08)	4.75 (0.13)

Table 3. Mean Labelers' Emotion Scores by Condition for the BFW Data set

Anchor	Bias mitigation	Happy	Angry	Neutral
Amazon	None	3.80 (0.10)	0.83 (0.04)	4.98 (0.09)
	Transparency	3.88 (0.10)	1.15 (0.05)	4.80 (0.09)
Microsoft	None	4.5 (0.10)	0.86 (0.04)	4.44 (0.09)
	Transparency	4.37 (0.11)	0.64 (0.04)	4.50 (0.10)
Control	None	4.27 (0.10)	0.81 (0.04)	4.62 (0.09)

demographic characteristics. Next, to shed light on both the anchoring effect and algorithmic interference inconsistencies, we rerun the analysis on only subjects that yielded such inconsistencies in the earlier part of our experiment.

5.2.3.1. NBA Data Set. We restrict the NBA data set to instances in which an emotion AI exhibited either false-positive inference inconsistency or false-negative inference inconsistency. This restriction leads to four different analysis data sets: Face++ and FNII; Face++ and FPII; Amazon and FNII; and Amazon and FPII. Microsoft does not produce inference inconsistencies.

Table 6 shows the results. The coefficient estimates on *Emotion AI – Happy* were not significant for the Face++ and FNII data set, but the coefficient estimates on *Emotion AI – Angry* were positive and significant ($\beta = 0.10$) for the Face++ and FPII data set. For Amazon, the coefficient estimates were positive and significant for *Emotion AI – Happy* and *Emotion AI – Angry* ($\beta = 0.06$ and $\beta = 0.69$, respectively) for the FNII and FPII data sets. These findings suggest that the anchoring effect is stronger for labelers' scores in the presence of FPII and in the presence of the inference inconsistencies from Amazon. The coefficient estimates on the anchors are positive and significant in the FPII tasks for both Face++ and Amazon ($\beta = 0.10$–0.05 and $\beta = 0.41$–0.69, respectively).

5.2.3.2. BFW Data Set. We do not report results for inference inconsistences in the BFW data set because neither Microsoft nor Amazon had any inference inconsistencies on this data set. In this data set, Amazon labeled nearly every face with the same emotion (calm), so either all the faces had inference inconsistencies or none did.

Table 4. Labelers' Scores for Happiness and Anger (NBA Data Set)

	Dependent variable					
	Happy			*Angry*		
Variable	(1)	(2)	(3)	(4)	(5)	(6)
Emotion AI – Happy	0.07*** (0.01)	0.07*** (0.01)	0.05** (0.02)			
Emotion AI – Angry				0.16*** (0.01)	0.16*** (0.01)	0.11*** (0.02)
African American		−0.14 (0.30)	−0.29 (0.32)		0.03 (0.13)	−0.01 (0.13)
African American × Emotion AI Score – Happy			0.03 (0.02)			
African American × Emotion AI Score – Angry						0.07* (0.03)
Constant	4.61*** (0.19)	4.71*** (0.30)	4.82*** (0.30)	0.80*** (0.14)	0.78*** (0.17)	0.81*** (0.17)
Observations	5,644	5,644	5,644	5,644	5,644	5,644
Log likelihood	−12,581	−12,582	−12,583	−10,230	−10,231	−10,231
Akaike information criterion	25,173	25,175	25,180	20,470	20,475	20,477

Notes. The dependent variable is the labelers' emotion score of *Happy* or *Angry* on a scale of 0–10. *Emotion AI – Happy* and *Emotion AI – Angry* are the emotion AI scores for the particular subject and condition on a scale of 0–10. *African American* is a binary indicator for whether the subject is African American. These coefficients are estimated using random effects for each image and emotion AI model. The data set comprises NBA images. Transparency conditions are excluded. Significance is indicated with asterisks.

*$p \le 0.05$; **$p \le 0.01$; ***$p \le 0.001$.

Table 5. Labelers' Scores for Happiness and Anger (BFW Data Set)

	Dependent variable			
	Happy		*Angry*	
Variable	(1)	(2)	(3)	(4)
Emotion AI – Happy	0.14***	0.16***		
	(0.01)	(0.02)		
Emotion AI – Angry			0.45***	0.52***
			(0.04)	(0.06)
Race – E. Asian	−0.30	−0.15	−0.18	−0.13
	(0.33)	(0.33)	(0.11)	(0.11)
Race – AA	0.41	0.54	−0.14	−0.12
	(0.33)	(0.33)	(0.11)	(0.11)
Race – S. Asian	−0.30	−0.30	−0.05	−0.03
	(0.33)	(0.33)	(0.11)	(0.11)
Emotion AI – Happy × Race – E. Asian		−0.05*		
		(0.02)		
Emotion AI – Happy × Race – AA		−0.04		
		(0.02)		
Emotion AI – Happy × Race – S. Asian		0.003		
		(0.02)		
Emotion AI – Angry × Race – E. Asian				−0.28*
				(0.14)
Emotion AI – Angry × Race – AA				−0.07
				(0.11)
Emotion AI – Angry × Race – S. Asian				−0.07
				(0.08)
Constant	3.77***	3.71***	0.82***	0.80***
	(0.24)	(0.25)	(0.09)	(0.09)
Observations	4,771	4,771	4,771	4,771
Log likelihood	−10,377.70	−10,382.71	−8,476.90	−8,479.01
Akaike information criterion	20,771.41	20,787.41	16,969.81	16,980.02

Notes. The dependent variable is the labelers' emotion score of *Happy* or *Angry* on a scale of 0–10. *Emotion AI – Happy* and *Emotion AI – Angry* are the emotion AI scores for the particular subject and condition on a scale of 0–10. *Race* is a set of binary indicators for whether the face is East Asian (E. Asian), African American (AA), or South Asian (S. Asian). These coefficients are estimated using random effects for each image and emotion AI model. The data set comprises BFW images. Transparency conditions are excluded. Significance is indicated with asterisks.

*$p \le 0.05$; ***$p \le 0.001$.

Face++ did indeed report inference inconsistencies in the data set, but Face++ was not used as an anchor; therefore, it cannot shed light on the anchoring effect.

These results are consistent with our expectations. Human labelers performing these tasks do display the anchoring effect, and this effect is moderated by the racial group of the subject, suggesting a selective anchoring effect. The anchoring effect appears weaker in instances with inference inconsistencies and easier tasks and stronger for instances with harder tasks, consistent with Hypotheses 1a and 1b.

5.3. Information Transparency and Inference Inconsistencies

We examine information transparency conditions to understand their effects on labelers' scores.

5.3.1. Information Transparency.

As discussed, we expect that when human labelers are provided with transparent information about AI fairness, it will lead to a weaker anchoring effect because the information about any disparities, or even the suggestion of disparities, will lead the labelers to reflect on their scores. We estimate a mixed-level model with the specification:

$$\begin{aligned} &LabelerScores_{ik} \\ &= \alpha + \beta_1 AIScore_i + \beta_2 Transparency_k + \beta_3 Race_i \\ &\quad + \beta_4 Transparency_k \times Race_i \\ &\quad + \beta_5 Transparency_k \times AIScore_i + \beta_6 Race_i \times AIScore_i \\ &\quad + \beta_7 Transparency_k \times Race_i \times AIScore_i + \epsilon_i, \end{aligned}$$

where *LabelerScores* are the labelers' scores (happiness or

Table 6. Labelers' Scores for Happiness and Anger for Subjects with Inference Inconsistences

	DV = Happy			*DV = Angry*			*DV = Happy*			*DV = Angry*		
	Face++ and FNII			Face++ and FPII			Amazon and FNII			Amazon and FPII		
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Emotion AI – Happy	0.06	0.06	−0.09				0.08***	0.08***	0.06*			
	(0.04)	(0.04)	(0.09)				(0.02)	(0.02)	(0.03)			
Emotion AI – Angry				0.10*	0.10*	0.05				0.41***	0.41***	0.69**
				(0.04)	(0.04)	(0.10)				(0.12)	(0.12)	(0.23)
African American		2.10*	1.31		−0.17	−0.21		−0.15	−0.32		−0.03	0.14
		(1.03)	(1.10)		(0.32)	(0.33)		(0.20)	(0.25)		(0.45)	(0.47)
African American ×			0.19						0.03			
Emotion AI – Happy			(0.10)						(0.02)			
African American ×						0.05						−0.31
Emotion AI – Angry						(0.10)						(0.25)
Constant	6.04***	4.29***	4.91***	0.70***	0.83**	0.86**	6.45***	6.57***	6.69***	2.10***	2.11***	1.97***
	(0.46)	(0.96)	(1.01)	(0.17)	(0.30)	(0.30)	(0.28)	(0.32)	(0.33)	(0.27)	(0.40)	(0.40)
Observations	287	287	287	624	624	624	2,503	2,503	2,503	406	406	406
AIC	1,324	1,320	1,321	2,181	2,183	2,188	11,183	11,186	11,192	1,820	1,821	1,823

Notes. These coefficients are estimated using random effects for each image and emotion AI model. The data set comprises NBA images. Transparency conditions are excluded. Significance is indicated with asterisks. AIC, Akaike information criterion; DV, dependent variable.
*$p \le 0.05$; **$p \le 0.01$; ***$p \le 0.001$.

anger) assigned to subject i in transparency condition k and *Transparency* is a binary indicator for whether the labeler receives information about AI fairness. For the NBA data set, *Race* is a binary variable for whether the subject i is African American. For the BFW data set, *Race* is a set of binary variables for whether person i is in a specific group (East Asian, African American, or South Asian). *Transparency* × *Race* is the interaction between the two variables. There is also a subject-level random effect ϵ_i to control for differences between subjects. For this analysis, we examine how human labelers react under the transparency condition for each emotion AI.

5.3.1.1. NBA Data Set.

Table 7 shows the regression results for the labelers' scores. Transparency is not significant for happiness in most conditions (columns (2)–(6)), but the coefficient estimates on *Transparency* are significant for anger in most conditions (columns (7)–(12)). The coefficient direction differs based on the emotion AI. Labelers exposed to anger scores generated by Face++ or Microsoft leave lower anger scores ($\beta = -0.10$ and $\beta = -0.15$), whereas those exposed to Amazon leave higher anger scores ($\beta = 0.21$). Therefore, there is some support for Hypotheses 2a and 2b. The three-way interaction is not significant, suggesting no evidence for Hypothesis 3a.

5.3.1.2. BFW Data Set.

Table 8 shows that the coefficient estimates for *Transparency* were mixed in significance for happiness (columns (1)–(4)) but mostly significant for anger (columns (5)–(8)). The coefficient estimates on *Transparency* were positive for exposure to Amazon's anger score ($\beta = 0.31$) and negative for exposure to Microsoft's anger score ($\beta = -0.24$). There is limited evidence of selective anchoring; the only significant estimate is the interaction between *Transparency* and *Race – African American* ($\beta = 0.31$), suggesting that transparency increased anger scores when labelers encountered African American faces. This analysis finds support for Hypotheses 2a and 2b but not for Hypothesis 3a.

5.3.2. Human Inference Inconsistencies.

The labelers' emotion scores provide insight into the anchoring effect but not necessarily into the quality of the labelers' scores. Like algorithms, human judgment can also produce inference inconsistencies. To gauge the quality of labelers' scores, we examine the effect of transparency on human inference inconsistencies and the percentage of labeler scores that exhibit inference inconsistency. For the NBA data set, the perceived smile is based on the labeler's perception; for the BFW data set, the perceived smile is based on the researcher's coding. Table 9 shows the labelers' inference inconsistency by emotion AI condition. We observe that human labelers make the same mistakes as the models, but there is evidence that human labelers are less likely to make those mistakes when they are provided with emotion AI scores. For the NBA data set, labelers show less FPII in all emotion AI conditions and less FNII in all emotion AI conditions except Face++. For the BFW data set, labelers show less FPII in all emotion AI conditions and less FNII in the Microsoft condition.

This preliminary analysis underlines the potential benefits of the anchoring effect, namely that providing human labelers with emotion AI scores could reduce inference inconsistencies among those labelers. Still,

Table 7. Labelers' Scores for Happiness and Anger with Information Transparency (NBA Data Set)

	Dependent variable = *Happy*						Dependent variable = *Angry*					
	FPP		AWS		MSFT		FPP		AWS		MSFT	
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Emotion AI – Happy	0.26***	0.32***	−0.04	−0.20*	0.69***	0.72***						
	(0.03)	(0.07)	(0.05)	(0.10)	(0.01)	(0.03)						
Emotion AI – Angry							0.17**	0.05	0.08***	0.06	−0.06	−0.08
							(0.06)	(0.18)	(0.02)	(0.05)	(0.39)	(0.55)
Transparency	0.23***	−0.26	0.08	0.004	−0.12	0.07	−0.10*	−0.24*	0.21***	0.38**	−0.15***	−0.15
	(0.07)	(0.25)	(0.07)	(0.18)	(0.07)	(0.25)	(0.05)	(0.10)	(0.06)	(0.13)	(0.04)	(0.08)
Race – African American	0.14	0.63	−0.28	−0.75	0.33**	0.50	−0.01	−0.07	0.10	0.01	−0.01	−0.01
	(0.29)	(0.55)	(0.34)	(0.41)	(0.13)	(0.27)	(0.13)	(0.15)	(0.14)	(0.17)	(0.10)	(0.11)
Emotion AI – Happy × African American		−0.10		0.24*		−0.03						
		(0.08)		(0.12)		(0.03)						
Emotion AI – Happy × Transparency		0.06		−0.002		−0.04						
		(0.03)		(0.05)		(0.03)						
Emotion AI – Angry × African American								0.11		0.09		−0.01
								(0.19)		(0.06)		(0.88)
Emotion AI – Angry × Transparency								0.37**		−0.06		0.11
								(0.14)		(0.04)		(0.46)
African American × Transparency		0.34		0.16		−0.12		0.17		−0.05		0.002
		(0.28)		(0.20)		(0.28)		(0.11)		(0.16)		(0.09)
Emotion AI – Happy × African American × Transparency		−0.03		−0.03		0.03						
		(0.04)		(0.06)		(0.04)						
Emotion AI – Angry × African American × Transparency								−0.37*		−0.04		−0.14
								(0.15)		(0.05)		(0.74)
Constant	3.08***	2.81***	5.19***	5.52***	0.66***	0.48*	0.80***	0.86***	0.67***	0.67***	0.64***	0.64***
	(0.32)	(0.49)	(0.31)	(0.36)	(0.14)	(0.24)	(0.12)	(0.13)	(0.13)	(0.15)	(0.09)	(0.10)
Observations	2,819	2,819	2,823	2,823	2,822	2,822	2,819	2,819	2,823	2,823	2,822	2,822
AIC	12,559	12,573	13,147	13,163	11,890	11,914	10,025	10,036	11,199	11,204	9,128	9,137

Notes. The dependent variable is the labelers' emotion score of *Happy* or *Angry* on a scale of 0–10. *Emotion AI – Happy* and *Emotion AI – Angry* are the emotion AI scores for the particular subject and condition on a scale of 0–10. *African American* is a binary indicator for whether the face is African American. These coefficients are estimated using random effects for each image. The data set comprises NBA images. Significance is indicated with asterisks. AIC, Akaike information criterion; AWS, Amazon; FPP, Face++; MSFT, Microsoft.

*$p \leq 0.05$; **$p \leq 0.01$; ***$p \leq 0.001$.

Table 8. Labelers' Scores for Happiness and Anger with Algorithmic Transparency (BFW Dataset)

	Dependent variable = *Happy* Scores				Dependent variable = *Angry* Scores			
	Amazon		Microsoft		Amazon		Microsoft	
Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Emotion AI – Happy (Anchor)	−2.86***	−4.50*	0.80***	0.75***				
	(0.72)	(1.75)	(0.01)	(0.02)				
Emotion AI – Angry (Anchor)					0.32***	0.30	0.90***	1.07***
					(0.09)	(0.21)	(0.05)	(0.08)
Transparency	0.08	0.02	−0.13*	−0.35*	0.31***	0.25*	−0.24***	−0.37***
	(0.06)	(0.16)	(0.05)	(0.15)	(0.05)	(0.12)	(0.04)	(0.09)
Race – East Asian	−0.28	−0.59	0.36**	0.26	−0.23	−0.13	−0.12	−0.17
	(0.35)	(0.45)	(0.13)	(0.21)	(0.13)	(0.19)	(0.10)	(0.12)
Race – African American	0.55	0.38	0.22	−0.19	−0.02	0.07	−0.09	−0.25*
	(0.35)	(0.46)	(0.13)	(0.22)	(0.13)	(0.20)	(0.10)	(0.12)
Race – South Asian	−0.14	−0.38	0.03	−0.06	−0.01	−0.11	−0.08	−0.04
	(0.35)	(0.47)	(0.13)	(0.22)	(0.13)	(0.20)	(0.10)	(0.12)
Emotion AI – Happy × Transparency		0.41		0.05*				
		(0.84)		(0.02)				
Emotion AI – Happy × Race – East Asian		2.65		0.03				
		(2.36)		(0.03)				
Emotion AI – Happy × Race – African American		1.59		0.07*				
		(2.22)		(0.03)				
Emotion AI – Happy × Race – South Asian		1.69		0.02				
		(2.20)		(0.03)				
Emotion AI – Angry × Transparency						0.24		−0.05
						(0.18)		(0.09)
Emotion AI – Angry × Race – East Asian						−0.18		0.44
						(0.31)		(0.39)
Emotion AI – Angry × Race – African American						−0.18		0.15
						(0.29)		(0.20)
Emotion AI – Angry × Race – South Asian						0.26		−0.41***
						(0.29)		(0.11)
Transparency × Race – East Asian		−0.01		−0.01		0.07		0.13
		(0.22)		(0.20)		(0.17)		(0.13)
Transparency × Race – African American		0.07		0.10		−0.02		0.31*
		(0.22)		(0.21)		(0.17)		(0.13)
Transparency × Race – South Asian		0.08		0.13		0.08		0.08
		(0.23)		(0.20)		(0.17)		(0.13)
Constant	4.16***	4.36***	0.60***	0.83***	0.76***	0.75***	0.82***	0.86***
	(0.26)	(0.33)	(0.11)	(0.16)	(0.11)	(0.14)	(0.08)	(0.09)
Emotion AI – Happy × Transparency × Race		Yes		Yes				
Emotion AI – Angry × Transparency × Race						Yes		Yes
Observations	3,143	3,143	3,143	3,143	3,143	3,143	3,143	3,143
AIC	14,515	14,524	12,096	12,143	11,684	11,713	11,1575	11,163

Notes. The dependent variable is the labelers' emotion score of *Happy* or *Angry* on a scale of 0–10. *Emotion AI – Happy* and *Emotion AI – Angry* are the emotion AI scores for the particular subject and condition. *Transparency* is a binary indicator for whether the disparities were reported. *Race* is a set of binary indicators for whether the face is East Asian, African American, or South Asian. These coefficients are estimated using the random intercept for each image. The data set comprises BFW images. Significance is indicated with asterisks. AIC, Akaike information criterion.
*$p \le 0.05$; **$p \le 0.01$; ***$p \le 0.001$.

false-negative rates can be higher when labelers are presented with photos of African Americans and South Asians, suggesting that the labeling quality is influenced by demographic characteristics. To further analyze these human inference inconsistences, we use the following model specification:

$$HumanInfIncons_{ik} = \alpha + \beta_1 Transparency_k + \beta_2 Race_i + \beta_3 Transparency_k \times Race_i + \epsilon_i,$$

where *HumanInfIncons* is a binary indicator for whether the labelers' score had an inference inconsistency, either FPII or FNII, for subject *i*. The other variables are similar to those defined. The data set is restricted to the anchor and transparency conditions to compare the information transparency on AI fairness within each emotion AI condition.

5.3.2.1. NBA Data Set.

Table 10 shows that information transparency can increase labelers' inference

inconsistencies. For Amazon and Microsoft emotion AI scores, the information transparency condition yields a significantly higher probability of FNII (columns (1) and (2) for Amazon and columns (3) and (4) for Microsoft). The coefficient estimates range from $\beta = 0.56$ to $\beta = 1.09$, suggesting that transparency may counterintuitively lead to lower-quality scores. In contrast, information transparency on AI fairness is mostly not significant for Face++ or for FPII (columns (5)–(12)). The interaction between *Transparency* and *African American* is significant for FNII and Amazon emotion AI exposure ($\beta = -0.69$); however, it is not consistent across conditions. Therefore, we find limited support for Hypothesis 3b.

Table 9. Labeler Inference Inconsistences

Data set, AI, and race	*N*	Human FNII, %	Human FPII, %
NBA			
None			
All	1,411	9.36	4.39
White	351	9.12	2.85
African American	1,060	9.43	4.91
Amazon			
All	1,412	6.80	2.20
White	351	4.56	3.70
African American	1,061	7.54	1.70
Face++			
All	1,409	11.30	2.56
White	349	12.60	2.58
African American	1,060	10.80	2.55
Microsoft			
All	1,412	5.74	2.05
White	350	5.14	1.71
African American	1,062	5.93	2.17
BFW			
None			
All	1,628	8.4	6.0
White	452	7.3	7.1
East Asian	392	7.7	8.4
African American	390	8.5	4.4
South Asian	394	10.4	4.1
Amazon			
All	1,572	11.5	3.8
White	396	11.1	3.8
East Asian	392	8.2	4.6
African American	390	11.0	3.1
South Asian	394	15.5	3.6
Microsoft			
All	1,572	6.2	6.2
White	396	5.6	6.8
East Asian	392	4.1	8.2
African American	390	6.4	4.1
South Asian	394	8.6	5.6

Notes. For labelers of the NBA data set, the percentage inconsistency is the percentage of people who label a face as smiling but do not label the emotion as happy. For the labelers of the BFW data set, the percentage inconsistency is the percentage of people who label a face as happy when Face++ says that the face is not smiling. Demographic parity conditions are excluded.

5.3.2.2. BFW Data Set.

Table 11 shows that the coefficients for transparency are mostly not significant. Similarly, there are significant, positive coefficient estimates on the interactions between *Transparency* × *Race – African American* and *Transparency* × *Race – South Asian* for FNII and Amazon exposure, suggesting that transparency associated with the extremely low emotion AI scores leads to poorer-quality decisions. Collectively, there is not support for Hypothesis 3b.

One confounding factor in this analysis is the dramatic variation in emotion AI scores, which may have inhibited labelers' ability to provide consistent labels. As one labeler commented about Amazon's emotion AI scores, "[a]lso as an aside you should rename your AI to AS, for artificial stupidity: p because it's rankings were very very off in my set." Despite the evidence for the anchoring effect, there is no evidence that labelers could meaningfully adjust for disparities or reduce their overall rate of inference inconsistency.

6. Discussion

In this paper, we find confirmation of the anchoring effect and a moderating effect of inference inconsistency, providing support for Hypotheses 1a and 1b. Information transparency regarding AI fairness is less influential when human labelers face easier emotion recognition tasks and stronger when they undertake harder emotion recognition tasks, consistent with Hypotheses 2a and 2b. This information transparency leads to spillover effects across all faces, with no differentiation based on the photo subject's demographic group. Because Hypothesis 3a hypothesized that information transparency about AI fairness would only affect scores for certain demographics, the results of this experiment do not demonstrate support for Hypothesis 3a and in fact, raise questions about labelers' ability to counteract AI disparities in their own scores. Plus, there is evidence that for certain faces, transparency increases the labelers' inconsistencies, providing limited evidence for Hypothesis 3b. Table 12 summarizes the support for each of the hypotheses.

This study makes multiple contributions to the information systems literature, and it builds on the algorithmic fairness literature by exploring the limits of information transparency on individuals' decisions. This paper suggests that even when individuals are provided with information transparency about AI fairness, it is not enough to help them overcome the anchoring effect and sufficiently adjust their behavior. However, this type of information transparency is still important; prior research has shown that information transparency about disparities can generate public pressure and/or regulations to reduce those disparities in the algorithm (Raji and Buolamwini 2019). These findings are consistent with those of other studies showing that the helpfulness of information transparency is

Table 10. Information Transparency and Inference Inconsistencies (NBA Dataset)

	Dependent variable = *Human FNII*					
	Amazon		Microsoft		Face++	
Variable	(1)	(2)	(3)	(4)	(5)	(6)
Transparency	0.56*** (0.14)	1.09*** (0.31)	0.60*** (0.15)	0.87** (0.30)	−0.08 (0.12)	−0.37 (0.25)
African American	0.11 (0.18)	0.55 (0.30)	−0.08 (0.19)	0.15 (0.30)	−0.001 (0.16)	−0.18 (0.21)
African American × *Transparency*		−0.69* (0.35)		−0.37 (0.35)		0.38 (0.29)
Constant	−2.96*** (0.20)	−3.31*** (0.28)	−3.05*** (0.21)	−3.22*** (0.28)	−2.23*** (0.16)	−2.10*** (0.18)
Observations	2,823	2,823	2,822	2,822	2,819	2,819
Log likelihood	−834.08	−832.04	−752.71	−752.13	−964.62	−963.73
AIC	1,676.16	1,674.07	1,513.42	1,514.26	1,937.24	1,937.45
	Dependent variable = *Human FPII*					
	Amazon		Microsoft		Face++	
Variable	(7)	(8)	(9)	(10)	(11)	(12)
Transparency	0.001 (0.26)	−0.41 (0.46)	−0.24 (0.31)	0.51 (0.59)	0.47* (0.22)	0.74 (0.42)
African American	−0.54 (0.36)	−0.84 (0.45)	−0.25 (0.59)	0.27 (0.71)	−0.25 (0.27)	−0.02 (0.41)
African American × *Transparency*		0.62 (0.56)		−1.05 (0.70)		−0.38 (0.49)
Constant	−4.58*** (0.51)	−4.40*** (0.53)	−6.94*** (1.01)	−7.39*** (1.07)	−3.88*** (0.34)	−4.06*** (0.42)
Observations	2,823	2,823	2,822	2,822	2,819	2,819
Log likelihood	−290.23	−289.61	−250.89	−249.74	−401.37	−401.08
AIC	588.47	589.23	509.78	509.48	810.75	812.16

Notes. The dependent variable is the binary variable for whether a labeler had an inference inconsistency for the subject. *Transparency* is a binary indicator for whether the demographic parity was reported to the labeler. *African American* is a binary indicator for whether the face is African American. These coefficients are estimated with random effects for image. The data set is comprised of NBA images. Significance is indicated with asterisks. AIC, Akaike information criterion.

*$p \le 0.05$; **$p \le 0.01$; ***$p \le 0.001$.

limited in the presence of bias. We, therefore, suggest that information transparency can merely change the locus of bias (Mejia and Parker 2021) rather than reduce it. Similarly, information transparency regarding AI fairness may lead to unintended consequences and is not a replacement for computational debiasing or higher-level oversight (e.g., regulations).

This study also indicates that cognitive biases limit the potential for individuals to counteract algorithmic bias. If the anchoring effect "corrupts" people's judgment on subjective tasks, such as consumer preferences and recommender systems (Adomavicius et al. 2013), human labelers may be so influenced by the AI output that they may not notice any bias in the AI. Labelers' scores exhibit inference inconsistencies that vary based on the demographic characteristics of the face, and this pattern is consistent with the anchoring effect's distorting influence on human discernment.

This study is also related to research on human-AI collaboration. Prior work along these lines has found that people rely more on AI advice for difficult tasks (Bogert et al. 2021). Thus, our observation that labelers' scores rely more on emotion AI for harder tasks is consistent with prior research. This study also suggests that providing human labelers with information about AI fairness is beneficial for such tasks.

This study also raises questions about emotion AI. Emotions are not necessarily universal or universally recognized (Elfenbein and Ambady 2002, Barrett 2017), and studies that depend on mapping facial expressions to emotions may not consider fairness issues (Choudhury et al. 2019). This study finds that artificial intelligence is erratic in its emotion interpretation, an observation consistent with those in other fields that question the universality of emotions (Gendron et al. 2014). In addition, we see that commercially available emotion AI models produce notably inconsistent scores for the same set of facial expressions and across demographic groups. Because of the variation in human interpretation and labeling of facial expression in emotions, these disparities are likely to persist, even as AI improves. This study provides additional evidence that AI tools, even those designed for the same task, are not necessarily interchangeable (Luca et al. 2016, Obermeyer et al. 2019). As emotion AI continues to become embedded in our communication technologies, it is important to consider its potential downsides. Many possible uses for emotion AI, such as measuring classroom engagement, have significant consequences for the human subjects. Although individuals, such as classroom moderators, may attempt to intercede and encourage fairness, these individuals may still experience

Table 11. Information Transparency and Inference Inconsistencies (BFW Data Set)

	Dependent variable							
	Human FNII				*Human FPII*			
	Amazon		Microsoft		Amazon		Microsoft	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Information Transparency	0.26	−0.68	0.30	0.41	0.64*	0.45	−0.07	−0.43
	(0.19)	(0.38)	(0.27)	(0.53)	(0.25)	(0.45)	(0.27)	(0.52)
Race – East Asian	−0.15	0.35	−0.33	−0.19	0.17	−0.12	0.33	1.24
	(0.65)	(0.59)	(1.01)	(1.11)	(0.67)	(0.67)	(0.82)	(0.92)
Race – African American	0.11	0.41	−0.05	0.40	−0.75	−0.86	−0.38	0.26
	(0.63)	(0.59)	(0.95)	(1.03)	(0.80)	(0.80)	(0.94)	(1.01)
Race – South Asian	0.41	−0.15	0.45	0.27	−0.01	0.59	−0.09	−0.10
	(0.61)	(0.60)	(0.87)	(0.98)	(0.68)	(0.63)	(0.88)	(1.08)
Transparency × Race – East Asian		0.99		−0.23		0.65		0.88
		(0.52)		(0.82)		(0.63)		(0.69)
Transparency × Race – African American		1.01*		−0.96		−0.37		0.70
		(0.51)		(0.80)		(0.78)		(0.80)
Transparency × Race – South Asian		1.35**		0.36		−0.53		−0.19
		(0.51)		(0.72)		(0.62)		(0.74)
Constant	−8.13***	−6.81***	−9.80***	−9.90***	−8.36***	−6.89***	−9.25***	−9.44***
	(0.57)	(0.54)	(0.80)	(0.85)	(0.67)	(0.61)	(0.74)	(0.91)
Observations	3,144	3,144	3,144	3,144	3,144	3,144	3,144	3,144
Log likelihood	−779.85	−796.09	−404.11	−402.55	−438.21	−445.20	−419.09	−420.21
AIC	1,571.69	1,610.18	820.22	823.09	888.43	908.39	850.19	858.41

Notes. The dependent variable is the binary variable for whether a labeler had an inference inconsistency for that face. *African American* is a binary indicator for whether the face is African American. These coefficients are estimated with random effects for image-level and emotion AI. The data set comprises BFW images. Significance is indicated with asterisks. AIC, Akaike information criterion.

*$p \le 0.05$; **$p \le 0.01$; ***$p \le 0.001$.

the anchoring effect, and thus, emotion AI scores may distort their judgment. The substantial inconsistency we observe in our experiment signals that we should reimagine the use of emotion AI in these spaces.

This study has multiple managerial implications. Organizations may adopt AI to scale their capabilities, such as searching for human talent (Tambe et al. 2019), and they may even apply AI specifically to help reduce bias (Cowgill 2017). However, extensive prior literature has found algorithmic bias to exist in contexts where it has serious consequences, such as hiring algorithms, criminal justice risk scores, facial recognition, and patient triage algorithms (Crawford 2013, Angwin et al. 2016, Barocas and Selbst 2016, Buolamwini and Gebru 2018, Obermeyer et al. 2019, Bartlett et al. 2022). In the United States, organizations are liable for any tool or policy that increases disparate outcomes along certain demographic characteristics, such as race or gender (Barocas and Selbst 2016). In this context, organizations that adopt third-party AI, such as emotion AI, cannot change the underlying algorithm and may wish to incorporate an interface to report aggregate algorithmic fairness measures to empower users. This study suggests that it may be unrealistic to expect

Table 12. Hypothesis Summary

Hypothesis	Support
The anchoring effect is weaker in the case of inference inconsistencies for easier tasks, such as mislabeled smiling faces (FNII) (Hypothesis 1a).	Supported
The anchoring effect is stronger in the case of inference inconsistencies for harder tasks, such as mislabeled nonsmiling faces (FPII) (Hypothesis 1b).	Supported
Information transparency on AI fairness will shape labelers' scores less in the easier task (Hypothesis 2a).	Supported
Information transparency on AI fairness will shape labelers' scores more in the harder task (Hypothesis 2b).	Supported
The effect of information transparency about AI fairness measures on labelers' scores is moderated by demographic characteristics (Hypothesis 3a).	None
Information transparency about AI fairness measures leads to inconsistency in labelers' scores (Hypothesis 3b).	Limited

individuals to produce fair outcomes through their own choices, even though fairness information is valuable in aggregate to hold AI developers accountable.

This paper has multiple limitations. First, we examine the anchoring effect and the effect of information transparency on AI fairness in an experimental context. This context provides a more controlled setting, but it would be interesting to examine this type of judgment in a real-world organizational setting. Second, NBA portraits are professional images with the expectations of similar facial expressions. Third, we use the emotion AI outputs rather than a random value as our anchor. Plus, this research assesses information transparency on AI fairness only in the context of emotion AI. Although it is beyond the scope for this manuscript, future research could explore if similar dynamics occur in other AI-related contexts.

To conclude, this paper underscores the limitations of information transparency about fairness measures as a decision support. In contexts in which organizations cannot change the underlying models, individuals can experience a strong anchoring effect during more difficult tasks, and even providing human users with information transparency about algorithmic fairness may still yield subpar results. Instead, we recognize that information about algorithmic fairness can be a valuable call to action.

Endnotes

[1] More formally, demographic parity is defined as $\mathrm{E}[\hat{Y}|A=a]-\mathrm{E}[\hat{Y}|A=0]=\hat{\mu}\leq\mu$, $a\in A$, where $\hat{Y}$ is the estimated outcome variable and A is the set of demographic characteristics. If the calculated disparity $\hat{\mu}$ is below a defined threshold μ, then the model is said to achieve fairness at the μ threshold.

[2] See https://www.faceplusplus.com/.

[3] See https://azure.microsoft.com/en-us/services/cognitive-services/face/.

[4] See https://aws.amazon.com/rekognition/.

[5] See https://www.technologyreview.com/2017/08/11/149962/when-a-face-is-worth-a-billion-dollars/.

[6] See https://www.basketball-reference.com/.

References

Adjerid I, Acquisti A, Loewenstein G (2018) Choice architecture, framing, and cascaded privacy choices. *Management Sci.* 65(5):2267–2290.

Adomavicius G, Bockstedt JC, Curley SP, Zhang J (2013) Do recommender systems manipulate consumer preferences? A study of anchoring effects. *Inform. Systems Res.* 24(4):956–975.

Angwin J, Larson J, Mattu S, Kirchner L (2016) Machine bias. *ProPublica* (May 23), https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing.

Ariely D, Loewenstein G, Prelec D (2003) "Coherent arbitrariness": Stable demand curves without stable preferences. *Quart. J. Econom.* 118(1):73–106.

Awwad Y, Fletcher R, Frey D, Gandhi A, Najafian M, Teodorescu M (2020) Exploring fairness in machine learning for international development. Technical report, CITE MIT D-Laboratory, Cambridge, MA.

Bänziger T, Grandjean D, Scherer KR (2009) Emotion recognition from expressions in face, voice, and body: The Multimodal Emotion Recognition Test (MERT). *Emotion* 9(5):691–704.

Barocas S, Selbst AD (2016) Big data's disparate impact. *Calif. Law Rev.* 104(3):671–732.

Barrett LF (2017) *How Emotions Are Made: The Secret Life of the Brain* (Houghton Mifflin Harcourt, Boston).

Bartlett R, Morse A, Stanton R, Wallace N (2022) Consumer-lending discrimination in the FinTech era. *J. Financial Econom.* 143(1):30–56.

Bassellier G, Ramaprasad J (2023). All external reference prices are not the same: How magnitude, source, and fairness shape payment for digital goods. *Inform. Systems Res.*, ePub ahead of print March 9, https://doi.org/10.1287/isre.2023.1206.

Bates D, Mächler M, Bolker B, Walker S (2015) Fitting linear mixed-effects models using lme4. *J. Statist. Software* 67(1):1–48.

Benitez-Quiroz CF, Srinivasan R, Martinez AM (2016) EmotioNet: An accurate, real-time algorithm for the automatic annotation of a million facial expressions in the wild. *2016 IEEE Conf. Comput. Vision Pattern Recognition (CVPR)* (IEEE, Piscataway, NJ), 5562–5570.

Bogert E, Schecter A, Watson RT (2021) Humans rely more on algorithms than social influence as a task becomes more difficult. *Sci. Rep.* 11(1):8028.

Brewer NT, Chapman GB (2002) The fragile basic anchoring effect. *J. Behav. Decision Making* 15(1):65–77.

Buolamwini J, Gebru T (2018) Gender shades: Intersectional accuracy disparities in commercial gender classification. *Proc. Machine Learn. Res.* 81:1–15.

Cao Z, Yan R, Huang Y, Shi Z (2019) Gigapixel-level image crowd counting using Csrnet. *2019 IEEE Internat. Conf. Multimedia & Expo Workshops (ICMEW)* (IEEE, Piscataway, NJ), 426–428.

Choudhury P, Wang D, Carlson NA, Khanna T (2019) Machine learning approaches to facial and text analysis: Discovering CEO oral communication styles. *Strategic Management J.* 40(11):1705–1732.

Cohen I, Garg A, Huang TS (2000) Emotion recognition from facial expressions using multilevel HMM. Leen TK, Dietterich TG, Tresp V, eds. *Neural Inform. Processing Systems*, vol. 2 (MIT Press, Cambridge, MA).

Corbett-Davies S, Pierson E, Feller A, Goel S, Huq A (2017) Algorithmic decision making and the cost of fairness. *Proc. 23rd ACM SIGKDD Internat. Conf. Knowledge Discovery Data Mining* (Association for Computing Machinery, New York), 797–806.

Cowgill B (2017) Bias and productivity in humans and algorithms: Theory and evidence from resume screening. Working paper, IZA world of labor, Washington DC.

Crawford K (2013) The hidden biases in big data. *Harvard Bus. Rev.* (April 1), https://hbr.org/2013/04/the-hidden-biases-in-big-data.

Dehghan A, Ortiz EG, Shu G, Masood SZ (2017) DAGER: Deep age, gender and emotion recognition using convolutional neural network. Preprint, submitted March 4, http://arxiv.org/abs/1702.04280.

Elfenbein HA, Ambady N (2002) On the universality and cultural specificity of emotion recognition: A meta-analysis. *Psych. Bull.* 128(2):203–235.

Eslami M, Kumaran SRK, Sandvig C, Karahalios K (2018) Communicating algorithmic process in online behavioral advertising. *Conf. Human Factors Comput. Systems Proc. 2018 April* (Association for Computing Machinery, New York), 1–13.

Fu R, Aseri M, Singh PV, Srinivasan K (2022) "Un"fair machine learning algorithms. *Management Sci.* 68(6):4173–4195.

Furnham A, Boo HC (2011) A literature review of the anchoring effect. *J. Socio-Economics* 40(1):35–42.

Gendron M, Roberson D, van der Vyver JM, Barrett LF (2014) Perceptions of emotion from facial expressions are not culturally universal: Evidence from a remote culture. *Emotion* 14(2):251–262.

Grother P, Ngan M, Hanaoka K (2019) Face recognition vendor test part 3: Demographic effects. NIST Interagency or Internal Report No. 8280, National Institute of Standards and Technology, Gaithersburg, MD.

Gunaratne J, Zalmanson L, Nov O (2018) The persuasive power of algorithmic and crowdsourced advice. *J. Management Inform. Systems* 35(4):1092–1120.

Halberstadt AG, Cooke AN, Garner PW, Hughes SA, Oertwig D, Neupert SD (2022) Racialized emotion recognition accuracy and anger bias of children's faces. *Emotion* 22(3):403–417.

Hardt M, Price E, Srebro N (2016) Equality of opportunity in supervised learning. Lee D, Sugiyama M, Luxburg U, Guyon I, Garnett R, eds. *Adv. Neural Inform. Processing Systems* (*NIPS*), vol. 29 (Curran Associates Inc., Red Hook, NY), 3315–3323.

Harwell D (2019) A face-scanning algorithm increasingly decides whether you deserve the job. *Washington Post* (October 22), https://www.washingtonpost.com/technology/2019/10/22/ai-hiring-face-scanning-algorithm-increasingly-decides-whether-you-deserve-job/.

Johnston PJ, Katsikitis M, Carr VJ (2001) A generalised deficit can account for problems in facial emotion recognition in schizophrenia. *Biological Psych.* 58(3):203–227.

Kilbride JE, Yarczower M (1983) Ethnic bias in the recognition of facial expressions. *J. Nonverbal Behav.* 8(1):27–41.

Kirita T, Endo M (1995) Happy face advantage in recognizing facial expressions. *Acta Psychologica* 89(2):149–163.

Kleinberg J (2018) Inherent trade-Offs in algorithmic fairness. *2018 ACM Internat. Conf. Measurement and Modeling of Computer Systems (SIGMETRICS '18)* (Association for Computing Machinery, New York), 40.

Kleinberg J, Ludwig J, Mullainathan S, Rambachan A (2018) Algorithmic fairness. *AEA Papers Proc.* 108:22–27.

Kotchemidova C (2005) Why we say "cheese": Producing the smile in snapshot photography. *Critical Stud. Media Comm.* 22(1):2–25.

Lehmann CA, Haubitz CB, Fügener A, Thonemann UW (2022) The risk of algorithm transparency: How algorithm complexity drives the effects on the use of advice. *Production Oper. Management* 31(9):3419–3434.

Leppänen JM, Hietanen JK (2004) Positive facial expressions are recognized faster than negative facial expressions, but why? *Psych. Res.* 69(1):22–29.

Logg JM, Minson JA, Moore DA (2019) Algorithm appreciation: People prefer algorithmic to human judgment. *Organ. Behav. Human Decision Processes* 151(2):90–103.

Luca M, Kleinberg J, Mullainathan S (2016) Algorithms need managers, too. *Harvard Bus. Rev.* 94(1):20.

Mehrabi N, Morstatter F, Saxena N, Lerman K, Galstyan A (2022) A survey on bias and fairness in machine learning. *ACM Comput. Surveys* 54(6):1–35.

Mejia J, Parker C (2021) When transparency fails: Bias and financial incentives in ridesharing platforms. *Management Sci.* 67(1): 166–184.

Mochon D, Frederick S (2013) Anchoring in sequential judgments. *Organ. Behav. Human Decision Processes* 122(1):69–79.

Morse L, Teodorescu MHM, Awwad Y, Kane GC (2022) Do the ends justify the means? Variation in the distributive and procedural fairness of machine learning algorithms. *J. Bus. Ethics* 181(4):1083–1095.

Nowicki S, Glanville D, Demertzis A (1998) A test of the ability to recognize emotion in the facial expressions of African American adults. *J. Black Psych.* 24(3):335–350.

Obermeyer Z, Powers B, Vogeli C, Mullainathan S (2019) Dissecting racial bias in an algorithm used to manage the health of populations. *Science* 366(6464):447–453.

O'Leary L (2022) What if Zoom could read your facial expression? *Slate* (May 16), https://slate.com/technology/2022/05/emotion-ai-zoom-sales-classroom.html.

O'Neil C (2016) *Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy* (Crown, New York).

Raji ID, Buolamwini J (2019) Actionable auditing: Investigating the impact of publicly naming biased performance results of commercial AI products. *Proc. 2019 AAAI/ACM Conf. AI Ethics Soc.* (Association for Computing Machinery, New York), 429–435.

Rhue L, Washington AL (2020) AI's wide open: Premature artificial intelligence and public policy. *Boston Univ. J. Sci. Tech. Law* 26(2):353–378.

Robinson JP, Livitz G, Henon Y, Qin C, Fu Y, Timoner S (2020) Face recognition: Too bias, or not too bias? *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition Workshops* (IEEE, Piscataway, NJ), 1–10.

Somers M (2019) Emotion AI, explained. *MIT Management Sloan* (March 8), https://mitsloan.mit.edu/ideas-made-to-matter/emotion-ai-explained.

Srivastava M, Heidari H, Krause A (2019) Mathematical notions vs. human perception of fairness: A descriptive approach to fairness for machine learning. *Proc. 25th ACM SIGKDD Internat. Conf. Knowledge Discovery Data Mining* (Association for Computing Machinery, New York), 2459–2468.

Tambe P, Cappelli P, Yakubovich V (2019) Artificial intelligence in human resources management: Challenges and a path forward. *Calif. Management Rev.* 61(4):15–42.

Telford T (2019) "Emotion detection" AI is a $20 billion industry. New research says it can't do what it claims. *Washington Post* (July 31), https://www.washingtonpost.com/business/2019/07/31/emotion-detection-ai-is-billion-industry-new-research-says-it-cant-do-what-it-claims/.

Teodorescu M, Morse L, Awwad Y, Kane G (2021) Failures of fairness in automation require a deeper understanding of human-ML augmentation. *MIS Quart.* 45(3):1483–1500.

Teodorescu MHM, Ordabayeva N, Kokkodis M, Unnam A, Aggarwal V (2022) Determining systematic differences in human graders for machine learning-based automated hiring. Working paper, Brookings Institute, Washington, DC.

Thomas D (2018) The cameras that know if you're happy—or a threat. *BBC News* (July 17), https://www.bbc.com/news/business-44799239.

Torralba A, Efros AA (2011) Unbiased look at data set bias. CVPR *2011* (IEEE, Piscataway, NJ), 1521–1528.

Turner Lee N, Resnick P, Barton G (2019) Algorithmic bias detection and mitigation: Best practices and policies to reduce consumer harms. *Brookings* (May 22), https://www.brookings.edu/research/algorithmic-bias-detection-and-mitigation-best-practices-and-policies-to-reduce-consumer-harms/.

Tversky A, Kahneman D (1974) Judgment under uncertainty: Heuristics and biases. *Science* 185(4157):1124–1131.

Vaish A, Carpenter M, Tomasello M (2009) Sympathy through affective perspective taking and its relation to prosocial behavior in toddlers. *Developmental Psych.* 45(2):534–543.

van Exel NJA, Brouwer WBF, van den Berg B, Koopmanschap MA (2006) With a little help from an anchor: Discussion and evidence of anchoring effects in contingent valuation. *J. Socio-Economics* 35(5):836–853.

Xu Y, Price T, Frahm J-M, Monrose F (2016) Virtual U: Defeating face liveness detection by building virtual models from your public photos. *25th USENIX Security Sympos. 4*, vol. 17 (ACM, New York), 497–512.

Xu T, White J, Kalkan S, Gunes H (2020) Investigating bias and fairness in facial expression recognition. Bartoli A, Fusiello A, eds. *Computer Vision – ECCV 2020 Workshops. ECCV 2020. Lecture Notes in Computer Science*, vol. 12540 (Springer, Cham), 506–523.

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